Using Piecewise-Linear Recurrent Neural Network to Predict the Vaping Behavior at the Next Prompt

Intro

Using vaporizers to vape nicotine is common among US young adults (ages 18-29 y), with 24.1% reporting vaping nicotine in the past month in 2023 [1,2]. Vaping nicotine is a growing public health concern because concurrent use has been associated with heavier substance use patterns, greater dependence severity, and increased exposure to harmful inhaled toxicants [3–5]. Young adulthood represents a critical period for intervention, as substance use behaviors often transition from experimentation to habitual use during this stage [6,7]. However, interventions specifically targeting nicotine vaping remain limited. To support more timely and personalized intervention strategies, it is important to identify real-time psychological and contextual predictors of near-term nicotine vaping risk.

The dataset used is from a smartphone-based ecological momentary assessment (EMA) study of vaping behaviors among young adults in United States. Eligible participants downloaded the study application to their personal smartphones and promoted to completed 4 EMA assessments a day at randomly scheduled times over a consecutive 30-day monitoring period. Each prompt remained active for 30 minutes, and survey completion typically required approximately 2–3 minutes. EMA assessments captured real-time information on vaping behavior, craving, affective state, and contextual factors. The study in total contained 130 participants. For each participant-level EMA observation, the outcome was derived from participants' responses to the question, "Are you about to vape right now?" If participants responded "yes," they were asked what type of vape they were about to use, including nicotine and cannabis. "About to vape" referred to the moment immediately before vaping initiation, when participants were preparing to use a vaping device. In this case, it can be interpreted as a proximal indicator of imminent vaping behavior. So, we will refer to the binary outcome as nicotine vaping. The main predictors include real-time nicotine craving, mood states, and contextual factors. The table 1 here summarizes the distribution of vaping outcomes and real-time predictors used in the PLRNN model.

Variable	All observations (n=11476)	Variable	All observations (n=11476)
Outcomes, n (%)		vape shop	22 (0.2)
Nicotine vaping / about to vape nicotine	3978 (34.7)	Time of day, n (%)	
Real-time predictors		Morning	2727 (23.8)
Personally staying with / social context, n (%)		Afternoon	4857 (42.3)
Alone	5824 (50.7)	Evening	3543 (30.9)
With people vaping nicotine	1009 (8.8)	Night	349 (3.0)
With people vaping cannabis	374 (3.3)	Day of week, n (%)	
With people smoking tobacco	191 (1.7)	1	1741 (15.2)
With people smoking cannabis	275 (2.4)	3	1723 (15.0)
Vaping allowed, n (%)		2	1665 (14.5)
Allowed	6056 (52.8)	0	1634 (14.2)
Forbidden	2862 (24.9)	4	1628 (14.2)
Discouraged	2257 (19.7)	5	1556 (13.6)
Don't know	301 (2.6)	6	1529 (13.3)
Location, n (%)		Craving and mood variables, mean (SD)	
home	6449 (56.2)	Nicotine craving	2.9 (1.8)
workplace	1869 (16.3)	Feeling happy	5.9 (2.1)
other place	1165 (10.2)	Feeling energized	5.4 (2.2)
in transit	1106 (9.6)	Feeling focused	5.3 (2.1)
friend's place	497 (4.3)	Feeling sad	2.8 (2.5)
school	259 (2.3)	Feeling stressed	3.6 (2.7)
bar or club	109 (0.9)	Feeling anxious	3.7 (2.7)

Method

For this project, I will use a classification-adapted piecewise-linear recurrent neural network, or PLRNN, to predict whether a participant will report vaping at the next ecological momentary assessment prompt. The outcome variable is binary: 1 indicates that the participant vape nicotine, and 0 indicates that the participant doesn't. However, the data are not ordinary independent observations. They are intensive longitudinal EMA data, meaning that each participant contributes repeated observations over time. Because craving, mood, context, and vaping risk may change dynamically across assessment, the PLRNN is appropriate because it models each participant's responses as a time series rather than as unrelated rows in a dataset. Another reason the PLRNN is appropriate is that it assumes an underlying latent state space, which can represent complex psychological dynamics that are not directly observed in the EMA responses.

The model separates variables into two groups. The first group is the dynamic EMA state, denoted by x_t. In this project, x_t includes nicotine craving, mood variables, and current about-to-vape status: x_t = {craving, happy, energized, focused, sad, stressed, anxious, current vape status}. The second group is the external or contextual input, denoted by u_t. These variables include time of day, day of week, location, social context, and whether vaping is allowed: u_t = {time of day, day of week, location, social context, vape allowed}. This distinction is important as the model treats craving and mood as part of the participant's internal behavioral state, while contextual variables are treated as external inputs that may impact how that state changes over time.

PLRNN first maps the observed EMA state into a latent state and then updates that latent state over time. The latent transition is modeled as: $z_{t+1} = Az_t + \Phi(z_t) + h + Cu_t$. In this equation, z_t is the latent state at time t. The term A z_t captures linear dependence on the previous latent state. The term h is a learned intercept or baseline drift term. The term Cu_t represents the influence of external contextual inputs. The nonlinear component is represented by $\Phi(z_t)$, which allows the model to capture state-dependent changes in the latent trajectory.

The nonlinear PLRNN component is: $\Phi(z_t) = W_1[\varphi(W_2 z_t + b) - \varphi(W_2 z_t)]$. Here, $\varphi(\cdot)$ is the ReLU activation function, which makes the model piecewise-linear. Because the outcome in this project is binary, I adapted the PLRNN into a classifier. After the latent state is updated, the model passes the next latent state through a classification layer: $\eta_{t+1} = w^T z_{t+1} + c$. In this equation, η_{t+1} is the logit for future about-to-vape status, w is a learned weight vector, and c is a learned intercept term. The logit is then transformed into a predicted probability using the sigmoid function: $P(y_{t+1} = 1|x_t, u_t) = \sigma(\eta_{t+1}) = 1/(1 + \exp(-\eta_{t+1}))$. The sigmoid function converts the logit into a probability between 0 and 1.

The model is trained using binary cross-entropy with logits, which is appropriate for a binary classification outcome. The loss function can be written as$L = -(1/N)\Sigma_{i=1}{}^{N}[y_i \log(p_i) + (1 - y_i)\log(1 - p_i)]$. In this equation, N is the number of training examples, y_i is the true label for observation i, and p_i is the predicted probability of future vape status. Because the outcome may be imbalanced, I use positive-class weighting so that errors on the less frequent class get greater weight during training.

Results

The EMA data were split using a within-participant temporal split. For each participant, observations were ordered by timestamp, with approximately 85% of observations assigned to the training set and the remaining 15% assigned to the testing set. Earlier observations were used for training and later observations were held out for testing to preserve the temporal structure of the data and evaluate whether the PLRNN could predict future nicotine vaping status from earlier EMA history.

The PLRNN was trained using binary cross-entropy with logits and positive-class weighting to account for imbalance in the vaping outcome. A validation set was used during training to select the best model. Specifically, validation loss was monitored after each epoch, and the model parameters from the epoch with the lowest validation loss were restored. Early stopping was used to reduce overfitting.

Classification results for predicting nicotine vaping status at the next EMA prompt are presented in Table 2. The training and testing set has similar nicotine vaping prevalence about 34%. On the testing set, the PLRNN achieved an accuracy of 0.767, precision of 0.657, sensitivity/recall of 0.659, specificity of 0.822, F1 score of 0.658, ROC-AUC of 0.774, and average precision of 0.623.

split	horizon	threshold	n	prevalence	accuracy	precision	sensitivity_recall	specificity	f1	tn	fp	fn	tp	roc_auc
train	step_1	0.5	8218	0.346556	0.731930	0.608696	0.634129	0.783799	0.621152	4209	1161	1042	1806	0.764444
test	step_1	0.5	1448	0.340470	0.766575	0.656566	0.659229	0.821990	0.657895	785	170	168	325	0.773778

Table 2. Classification performance of the PLRNN model for predicting future nicotine vaping status at the next EMA prompt.

Overall, the testing results suggest that the PLRNN had moderate ability to distinguish future nicotine vaping moments from non-vaping moments. The model showed stronger specificity than sensitivity, meaning it was better at correctly identifying non-vaping moments than detecting all vaping-risk moments. Testing performance was slightly higher than training performance across several metrics, which does not suggest obvious overfitting in this temporal split. These findings suggest that temporal patterns in craving, mood, prior vaping status, and context may provide useful information for predicting near-term nicotine vaping risk. Future analyses could extend this work by using the trained PLRNN to derive ideographic networks to visualize dynamic connections among craving, mood, context, and vaping behavior. Also, future work should compare PLRNN performance with simpler baseline models to determine whether the added model complexity provides meaningful predictive improvement.